Valuation of other financial securities

Bonds (Loan stocks/notes)

Features of a bond

- Is a debt instrument issued by either government (Government bond) or a corporate entity (corporate bond) when trying to borrow from the public.
- Is essentially a loan with pre-specified terms and conditions (for issuer and buyer) contained in the bond indenture (legal contract).
- Are negotiable debt instruments (can be traded)
- An instrument of indebtedness of the bond issuer to holders

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Features of a bond cont.

- *Par value/face value/nominal value:*
  - Amount the bond issuer promises to pay at maturity.
- *Coupon rate:*
  - Interest rate attached to the bond.
- *Coupon payment :*
  - Interest payable on the bond calculated as; coupon rate x par
- *Time to maturity:*
  - Length of time before the bond matures and is due for redemption.
- *Redemption value:*
  - Value at which the bond is bought back by issuer (usually at nominal value unless otherwise stated)

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Bond pricing

- The value of any financial security is the sum of the present values of all future cash flows associated with a particular financial asset.
- Thus, the value of any bond is the sum of the present values of all future cash flows associated with a particular bond.
- The discount rate used is the market interest rate.
- The market interest rate is inversely proportional to the bond price. i.e. bond price rises with a decline in market interest rate.

Bond pricing; *Zero coupon bond*

- **The only cash flow is the redemption value which often times is the face value.**
- **Discount rate is the current market interest rate.**
  - **Value of zero-coupon bond (Po) = $\frac{\text{Face value}}{(1 + r)^n}$**

Where : r = market interest rate

n = Time to maturity

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Activity 1

- **Z Co has in issue zero coupon K1,000 bonds. The bonds are redeemable at par in 3 years time. The current market interest rate is 8% per annum.**
- **<u>Required:</u>**

a) **Calculate the market price of the bond if interest is charged once a year.**

b) **Calculate the market price of the bond if interest is charged semi-annually.**

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Solution 1

a) Po = $\frac{K1000}{(1.08)^{\wedge 3}}$	K793.83

b) Interest is charged semi annually	Interest = 0.08/2 =0.04
Period will be 3 * 2 = 6 periods	
Po = $\frac{K1000}{(1.04)^{\wedge 6}}$ = K790.31	

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Bond pricing; Perpetual (irredeemable) bonds

- The only cash flow is the coupon payments which forms a perpetuity.
- Discount rate is the current market interest rate.

❖ Value of perpetual coupon bond (Po) = $\frac{\text{coupon payment}}{r}$

Where : r = market interest rate

Coupon payment = coupon rate x par value

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Activity 2

- **P Co has issued 9% K1,000 irredeemable bonds. The current market interest rate is 8% per annum. Company tax is 25%.**
- **<u>Required:</u>**

a) **Calculate the market price of the bond (ignore tax)**

b) **Calculate the market price of the bond using the interest net of tax.**

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Solution 2

a) **Coupon payment = 9%*K1000 = K90**

Po = $\frac{K90}{0.08}$ = *K1,125*

b) **Coupon payment net of tax = K90*(1-0.25) = K67.50**

Market interest net of tax = 8%*(1-0.25) = 6%

Po = $\frac{K67.50}{0.06}$ **= K1,125**

Bond pricing; Redeemable bonds

- The cash flows are the coupon payments and the redemption value.

- Bond price (Po) = $\frac{C}{(1+r)^1} + \frac{C}{(1+r)^2} \ldots + \frac{C+F}{(1+r)^n}$

Where : r = market interest rate

C = Interest (Coupon) payments

n = time to maturity

F = redemption value (face value).

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Activity 3

- **Q Co has issued 7% K100 redeemable bonds. These bonds are redeemable at par in 3 years time. The current market interest rate is 8% per annum.** ***C = 7%*K100 = K7 per year for 3 years, n = 3, F = K100, r =8% = 0.08***
- **<u>Required:</u>**

a) **Calculate the market price of the bond.**

b) **Calculate the market price of the bond if market interest rate falls to; (i) 7% per annum or (ii) falls to 6% per annum.**

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Solution 3

Coupon payment per year = 7%* K100 = K7 for 3 years

Redemption value = K100 in year 3

a) If market rate is 8%,

$$Po = \frac{7}{1.08^1} + \frac{7}{1.08^2} + \frac{(7+100)}{1.08^3} = 6.48 + 6.00 + 84.94 = \underline{K97.42}$$

b)i) ***if market interest falls to 7%***

$$Po = \frac{7}{1.07^1} + \frac{7}{1.07^2} + \frac{(7+100)}{1.07^3} = 6.542 + 6.114 + 87.344 = \underline{K100}$$

b)ii) ***if market interest falls to 6%***

$$Po = \frac{7}{1.06^1} + \frac{7}{1.06^2} + \frac{(7+100)}{1.06^3} = 6.60 + 6.23 + 89.84 = \underline{K102.67}$$

Practice question 1

- **ABC Co has issued 9% K1,000 bonds redeemable at a 5% premium to the par value in 3 years time. Company tax is 20% while the interest in the market net of tax is 10%.**
- **<u>Required:</u>**

Calculate the market price of the bond.

- **Market price = K72*annuity factor (10%, 3yrs) + K1,050*PVIf 10%,3yrs**
- **= (72*2.487) + (1050*0.751)**
- **= 179.06 + 788.55**
- **=<u>K967.61</u>**

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Solution 3

Coupon payment per year = 9%* K1,000 = K90 per for 3 years

Coupon payment net of tax = K90*(1-0.2) = K72 per year for 3 years

Redemption value at a 5% premium = K1,000 * 1.05 = K1,050 in year 3

Discount = 10%

$$Po = \frac{72}{1.1^1} + \frac{72}{1.1^2} + \frac{(72 + 1{,}050)}{1.1^3} = \underline{K967.61}$$

Alternatively

Year	Cash flows	Df at 10% $1/(1+r)^n$	PVs
1	72	0.909	65.448
2	72	0.826	59.472
3	1,122	0.751	842.622
		Market price	967.542

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Bond price and interest rates

- ***There is an inverse relationship between the market price of the bond and the current market interest rates.***
- **Increase in market interest rate reduces the bond price**
- **Reduction in the market interest rates increases the current bond price.**
- **Coupon rates (and coupon payments) do not change with the change in market interest rate (fixed-income security).**
- **Current market interest will be the coupon rate for a similar bond issued today.**

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Yield to Maturity (YTM)

- **Yield to maturity(YTM) is the market interest rate that equates the present values of the bond's future cash flows to its current market price.**
- **YTM is the internal rate of return (IRR) of the bonds promised cash flows (a rate at which the bond breaks even)**

$$\text{YTM (IRR)} = A + \frac{a}{(a-b)} \times (B - A)$$

Where:

A= Lower rate of interest with a positive NPV (a)

B = Higher rate of interest with a negative NPV (b)

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Activity 4

- **TC Co has issued 9% K100 redeemable bonds currently trading at K98.25 per K100 nominal. The bonds are redeemable at par in 3 years time.**
- **<u>Required:</u>**
- **Calculate the bonds yield to maturity.**

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Solution 4

Year	Cash flow K	Discount at 9%	Present Value (K)	Discount at 10%	Present value (K)
0	(98.25)	1.000	(98.25)	1.000	(98.25)
1	9	0.917	8.25	0.909	8.18
2	9	0.842	7.58	0.826	7.43
3	109	0.772	84.15	0.751	81.86
NPV			*1.73*		*(0.78)*

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Solution 4 cont.

- A = 9%, a = K1.73, B= 10% , b = (0.78)
- IRR (YTM) = 9% + {(1.73/(1.73+0.78))*(10%-9%)
- = 9% + 0.68%
- = 9.68%

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Bond duration

- **Bond duration is the sensitivity of the bond price to changes in interest rates.**
- **It shows the rate of change of the bond price given the change in interest rates (yields).**
- **The number of period in the future where the bond's average value is realized.**
- **The greater the duration, the farther into the future its value will be generated, the more volatile is the bond price to changes in interest rates.**

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Bond duration

- **Duration is calculated as = $\frac{\Sigma \text{ (PV of CF x t)}}{\text{Bond price}}$**

  <u>Where:</u> t = specific time when cash flow is generated

  PV of CF = present value of specific cash flows

- **Bond duration is measured as a period of time.**
- **It shows the payback period of the bond's cash flows.**

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Practice question 2

- **Calculate the bond duration of the following bonds given that market interest rate is 6% per year.**
- **Bond A: 10% K1,000 bond redeemable at par in 4 years times.**
- **Bond X: 12% K1,000 redeemable at a 2% premium in 2 years time.**
- **Bond Z: K1,000 zero coupon bond redeemable at par in 5 years time.**

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Activity 6

ERS has issued 10% K1,000, 4 year bonds . The bonds are redeemable at their nominal value while interest is paid annually. The current market interest rate is 6%.

<u>Required:</u>

a) **Calculate the bond's current market price and bond duration**

b) **Determine the percentage change in bond's price if the interest rate declined by 200 basis points.**

- **Hint: 100 basis points = 1%; 1 basis point = 0.01%**

Solution 6

Year (t)	Cash flow K	Discount at 6%	Present Value (K)	PV *t
1	100	0.943	94.3	94.30
2	100	0.890	89.0	178.00
3	100	0.840	84.0	252.00
4	1100	0.792	871.20	3,484.80
		Sum of PVs	*1,138.50*	*4,009.10*

Solution 6 cont.

- **Duration is** $= \dfrac{\Sigma \text{ PV of CF x t}}{\text{Bond price}}$
- $= 4{,}009.10/1{,}138.50$
- $= 3.52$ **years**
- **This means that it will take approximately 3.52 years for an investor in this instrument to recoup their investment.**
- **Similarly the results indicates that the price/value of this bond will change by 3.52 for every 1% change in the market interest rate.**

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